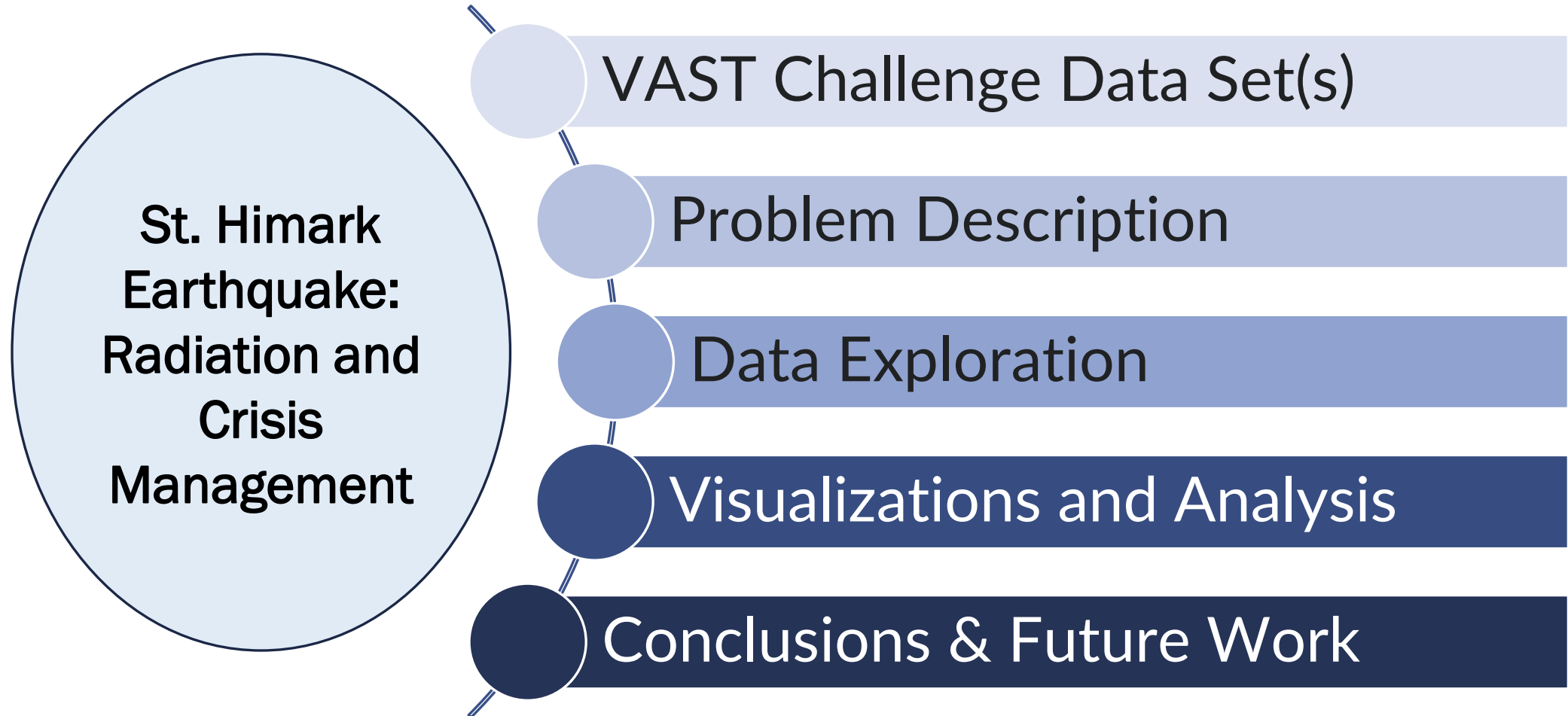




Crisis Mapping: St. Himark 2019

GROUP 2: CHARLES DEVLEN, SARAH GOTHARD, KONAIN NIAZ, KENNA PEZZI

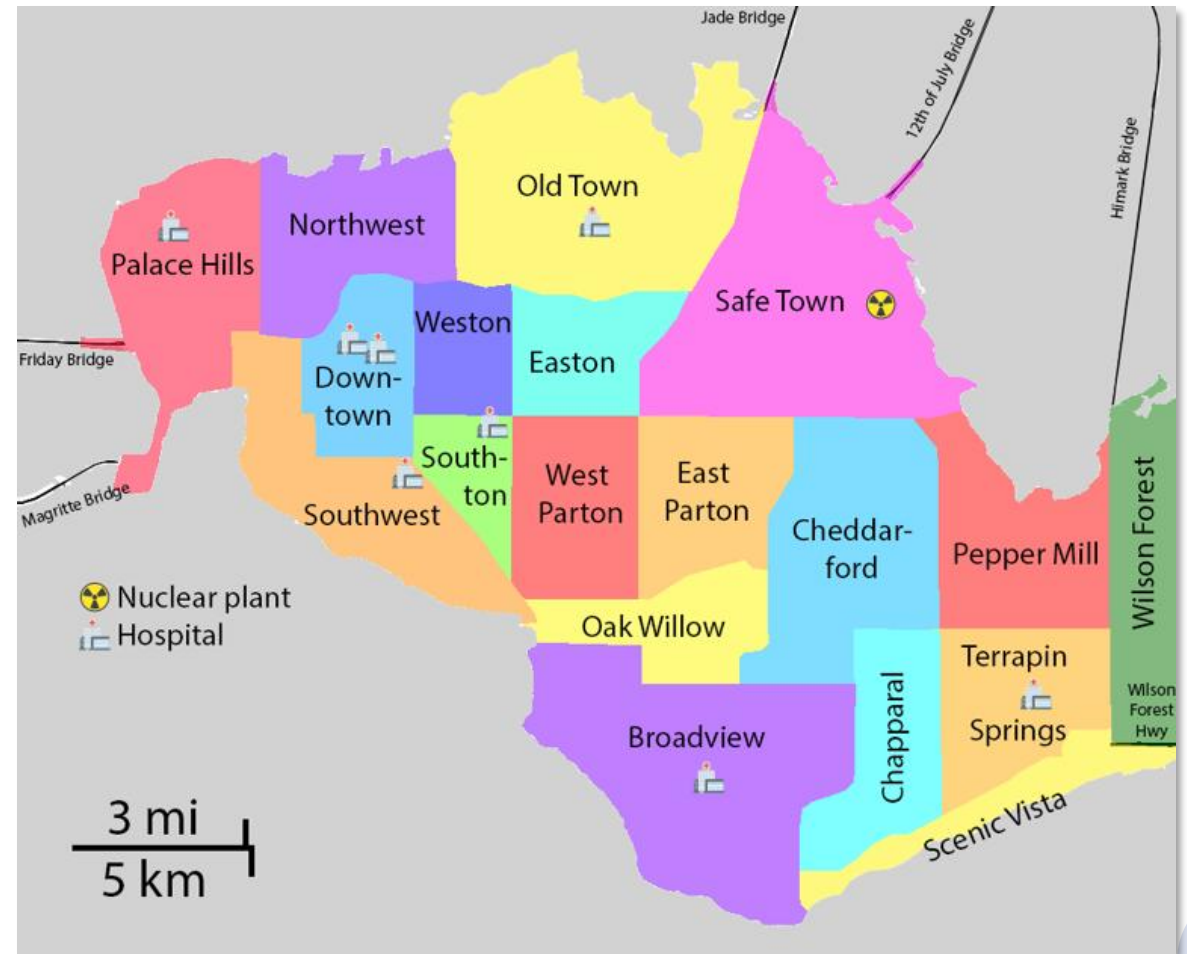
Presentation Agenda



DATA SET: EARTHQUAKES AND DISASTERS



- The VAST 2019 dataset was created around a fictitious disaster following an earthquake in the city of St. Himark
- The data was collected over a period of 5 days, from April 6th to April 10th.
- 2 data sets:
 - Mobile Sensors (3,315,711 records): Citizen scientist sensors
 - Static Sensors (744,000): fixed place calibrated sensors
- 7 Original features (timestamp, sensor_id, longitude, latitude, radiation value, units, user_id)
- Engineered features (Hours, Date, Weekday, Sensor Type [binary between mobile/static])
- Shape file maps





PROBLEM DESCRIPTION:

After the earthquake, St. Himark's emergency management team needs to better understand conditions throughout the city and identify areas to be monitored:

- Address uncertainty in reports from the citizen scientists with mobile sensors
 - Report on model variation
 - Analyze conflicting reports/information through mean evaluation
- Troubled area identification
 - Examine the spread of contamination over time
 - Determine which regions of the city need the most attention
- Make plans for cleanup and evacuation
 - Create an animated map of St. Himark with radiation levels shown from mobile and fixed sensors over time



Data Exploration



Descriptive Statistics Interesting Findings:

- Sensor *Labeling had 9 Overlapping names* between Mobile Data Set and Static
- There was *0 Intersection of Location* (long/lat) between both data sets
- Radiation *Values Were Distinctly Higher for Mobile Sensor Reports* than the Static Sensors
- Mobile Sensor had 1 extreme *outlier of a CPM value of 57,345.2* (user CitizenScientist) – *removed for analysis*

Table One

	Mobile Sensor (n=3,315,711)	Static Sensor (n=744,000)
	Mean(SD)	Mean(SD)
Longitude	-119.85(0.058)	-119.85 (0.056)
Latitude	0.127 (0.048)	0.145 (0.046)
Value	45.98 (161.03)	15.79 (11.88)
Hour	11.14 (7.07)	11.51 (6.93)
	Unique Counts	Unique Counts
Sensor ID's	50	9
Units	1	1
User Id's	22	1
Week Day	5	5

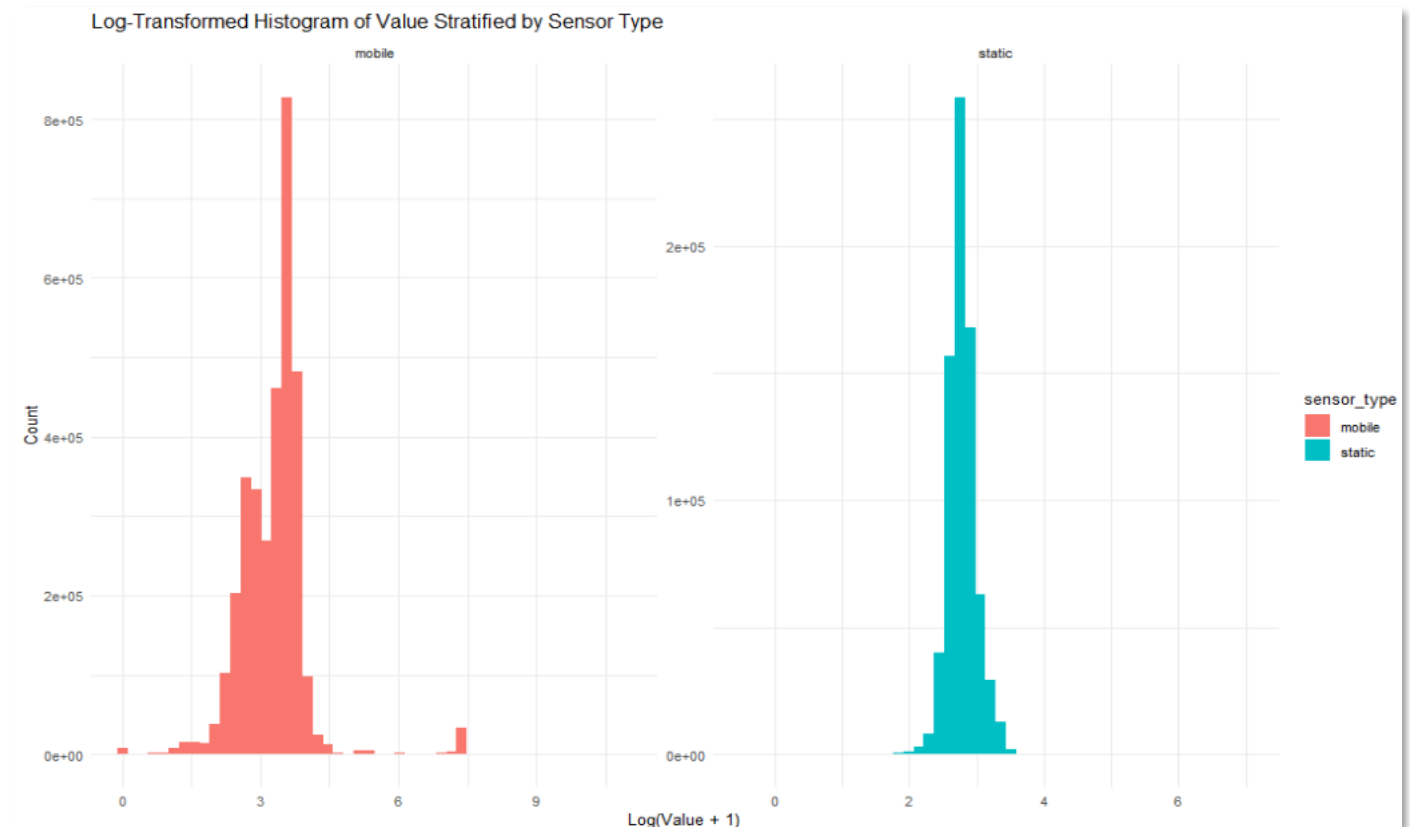


Data Exploration



VALUE Variable

- Heavy *untransformed skew to right* (most values <500 mobile, <100 static)
- *Log transformation* shows very narrow, tall distribution appearing *approximately normal*

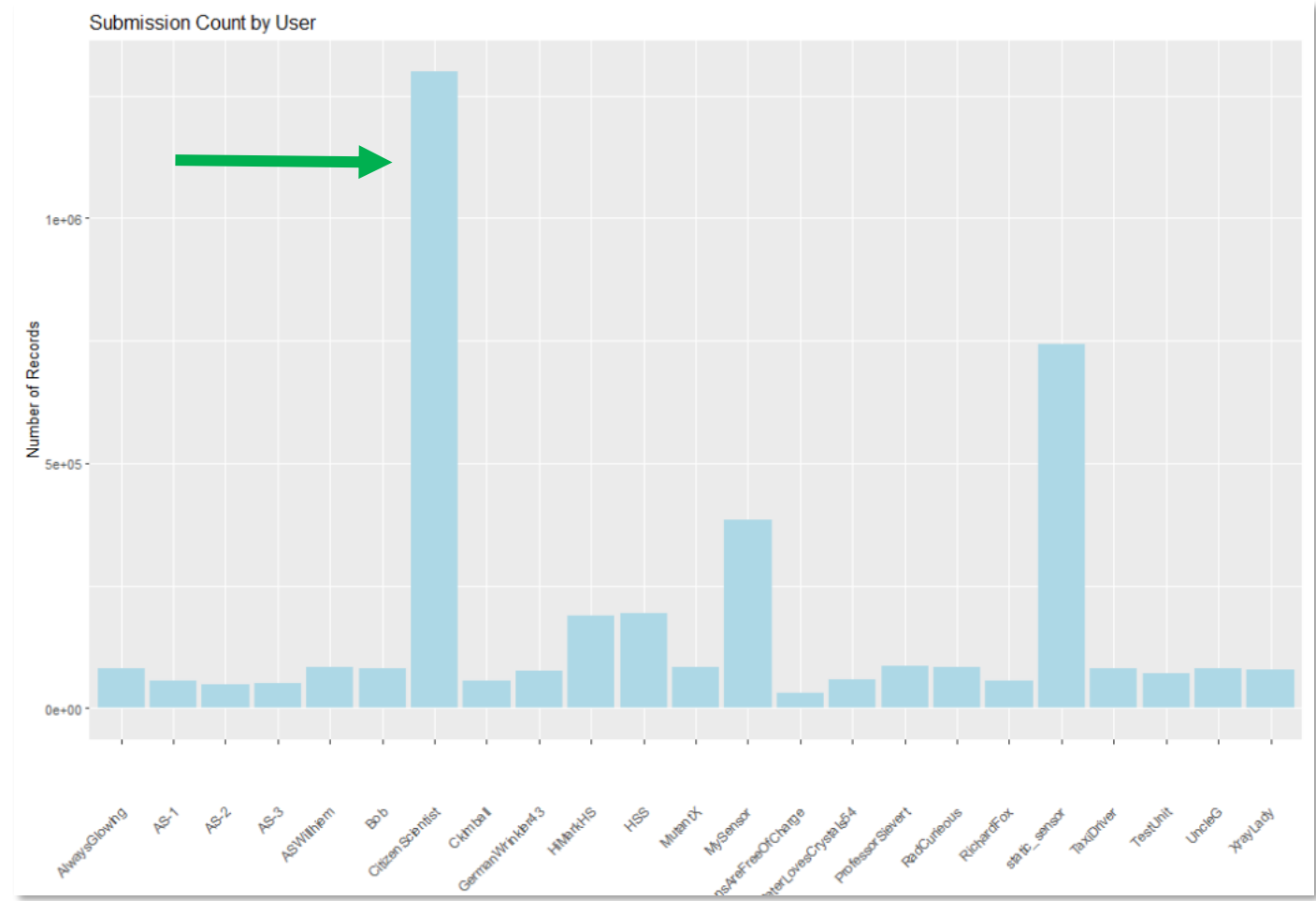


Data Exploration



User Counts

- Citizen Scientist was *overly represented* in the count distribution of the mobile users.

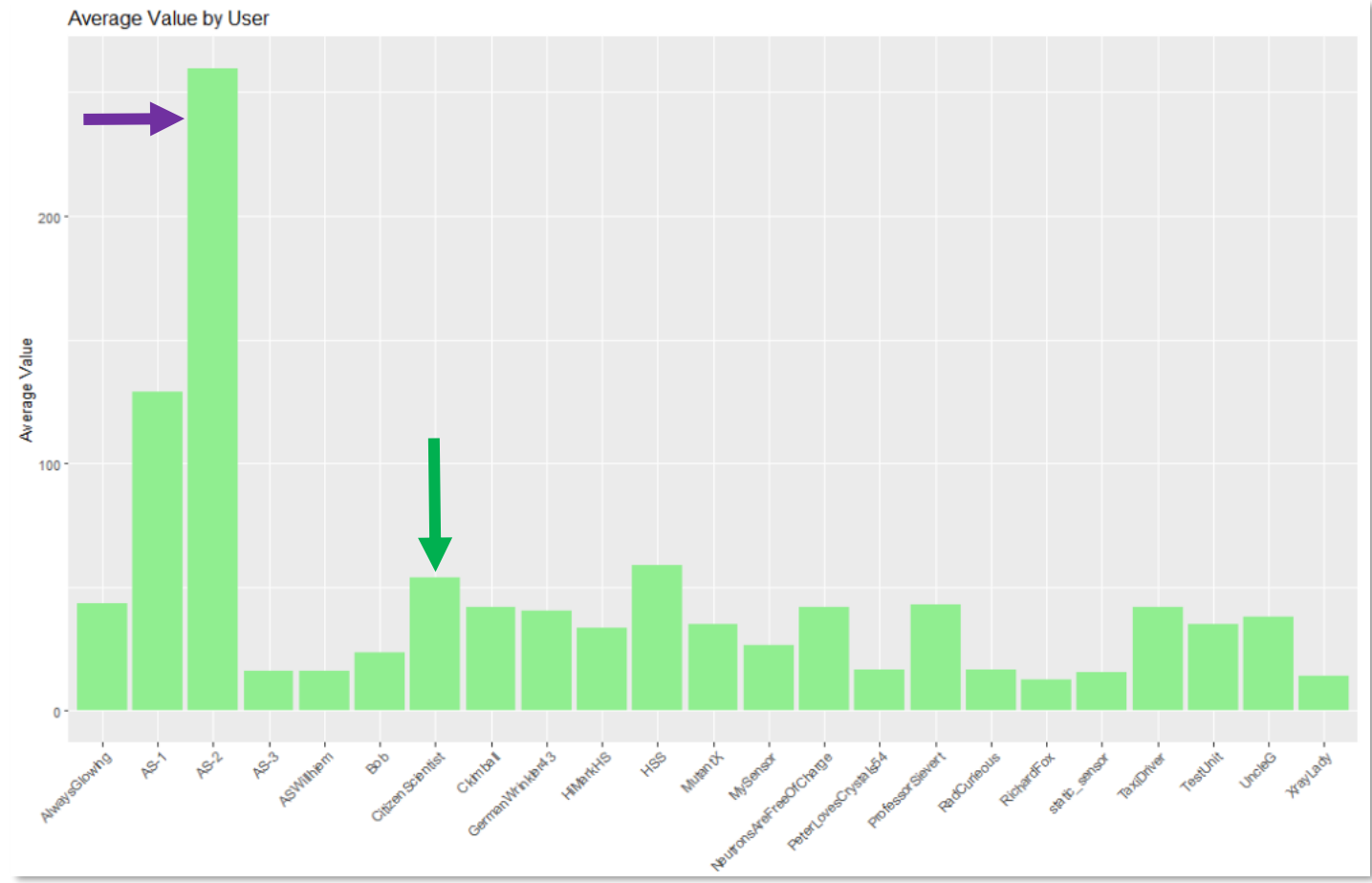


Data Exploration



User Averages

- **Citizen Scientist** was overly represented in the count but **AS-2** reported the highest average values

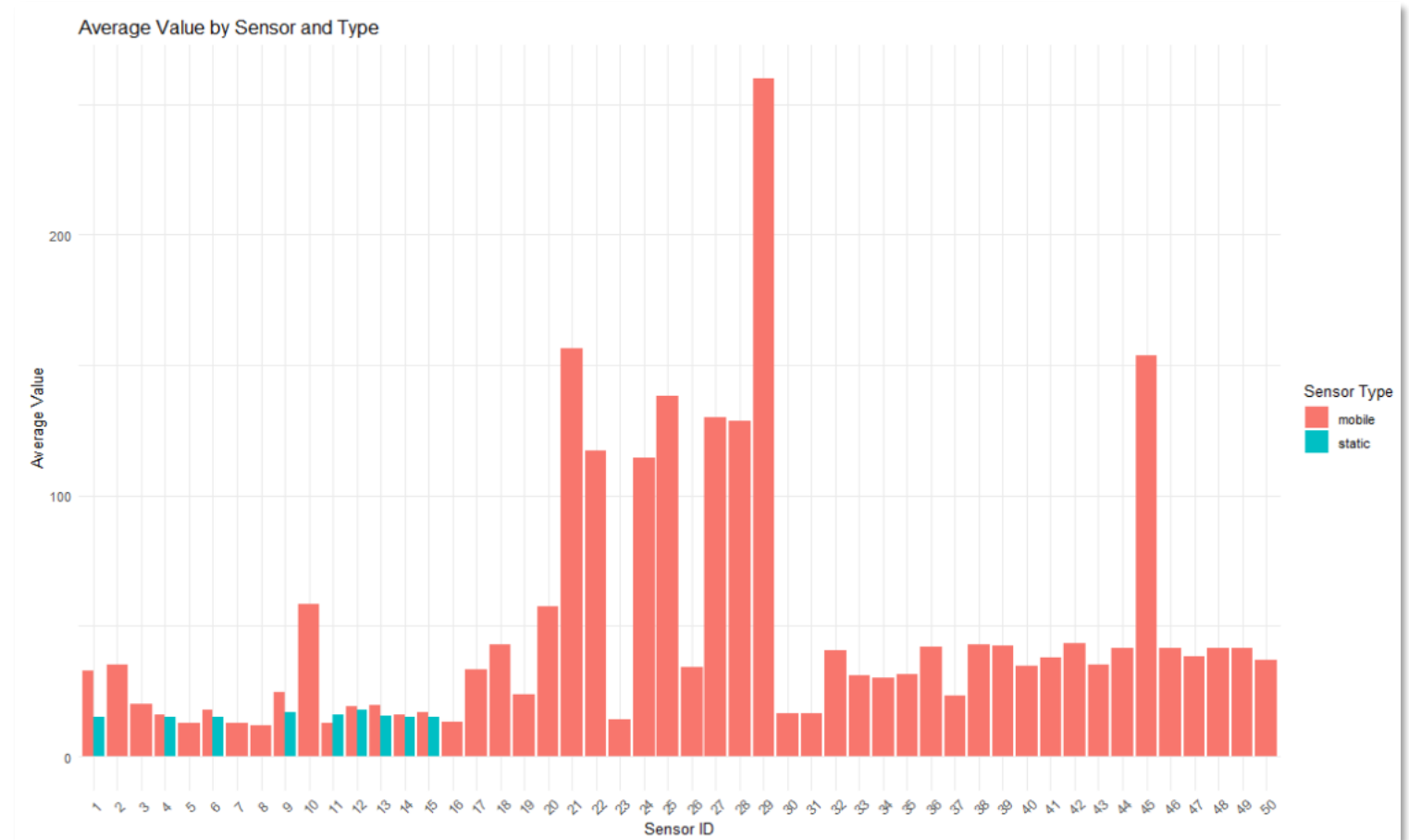


Data Exploration



Average Values By Sensor

- Sensors from Mobile
User's reported high CPM values

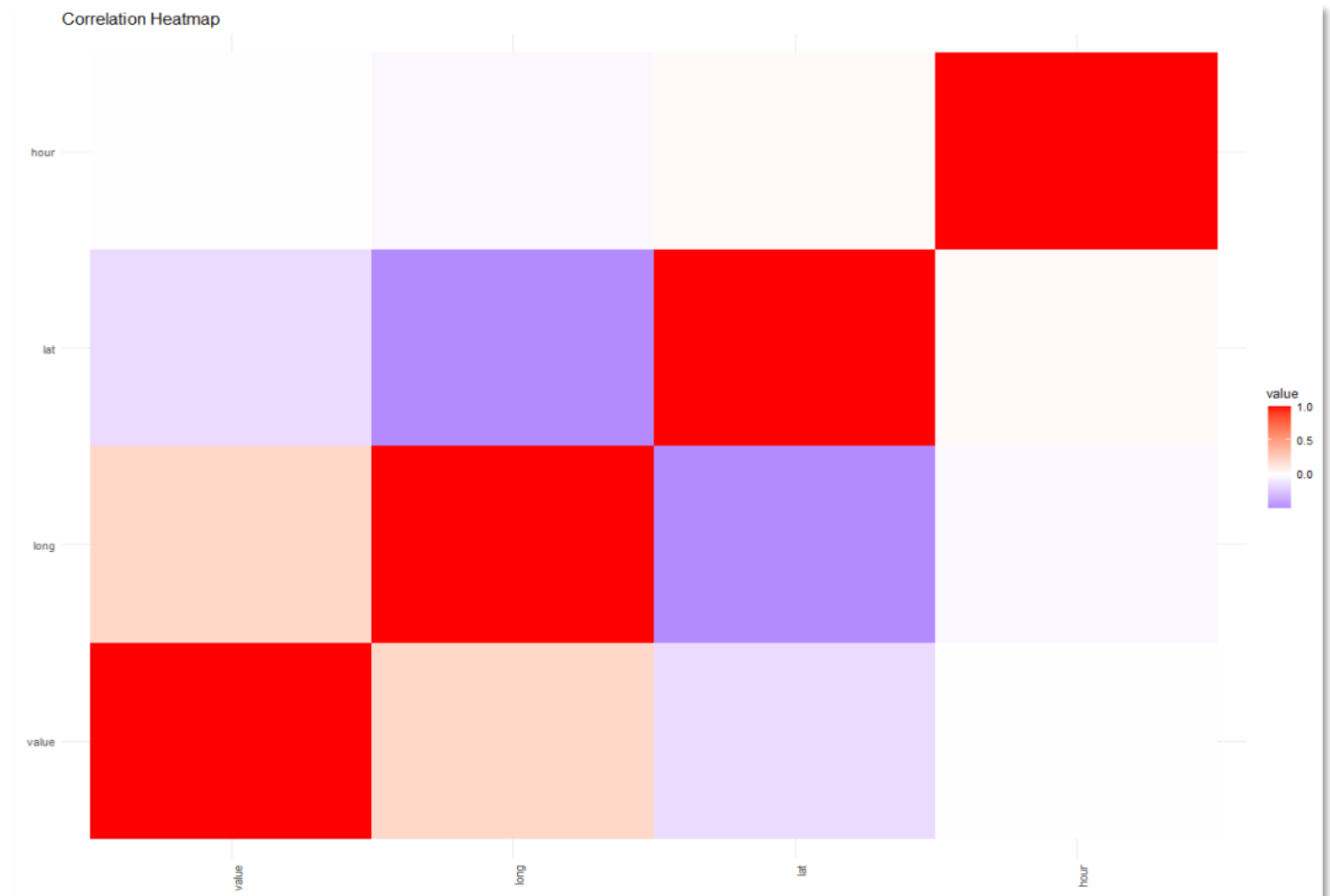


Data Exploration



Correlation

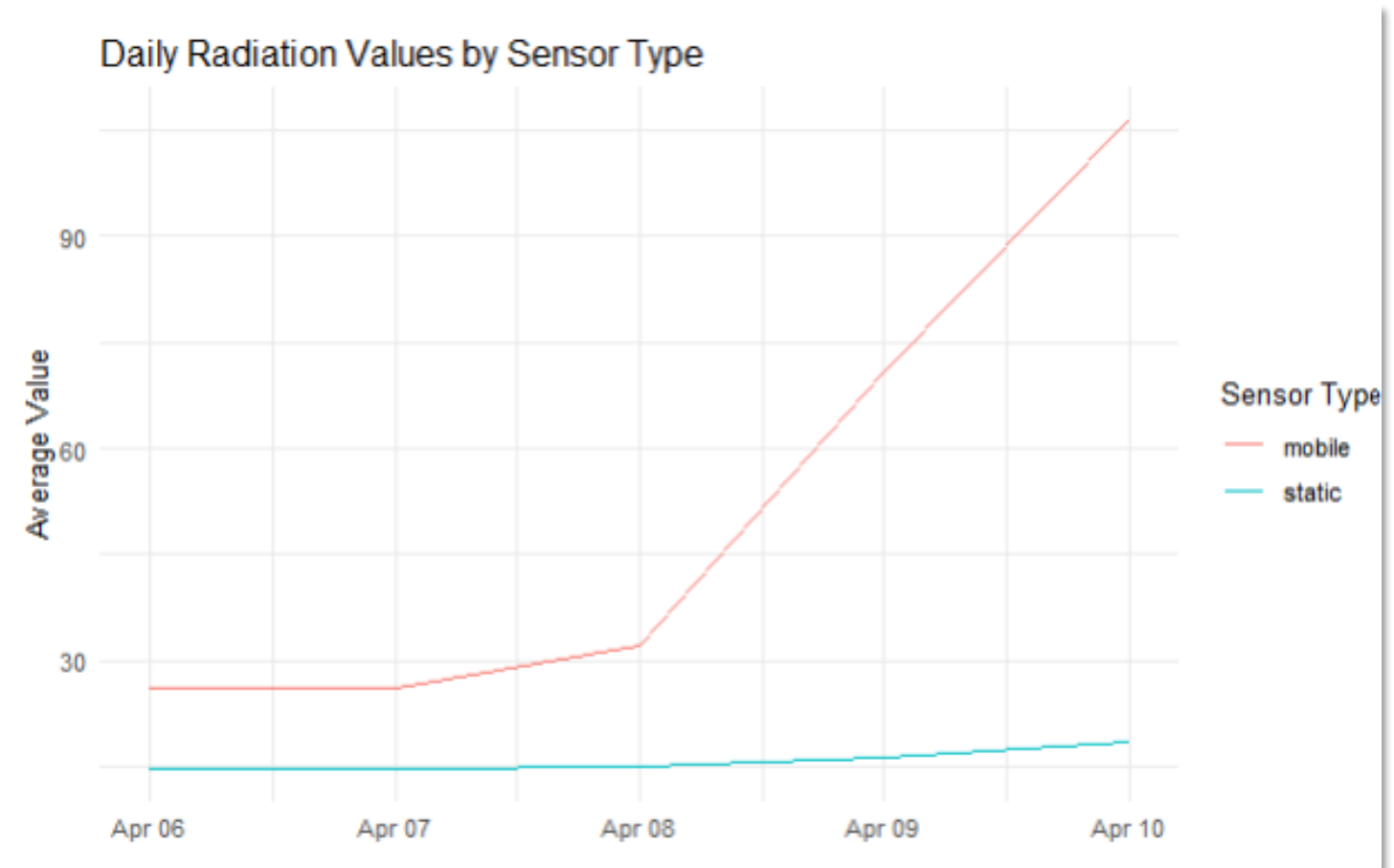
- For aggregate data, there was a moderate correlation between longitude and the value of radiation



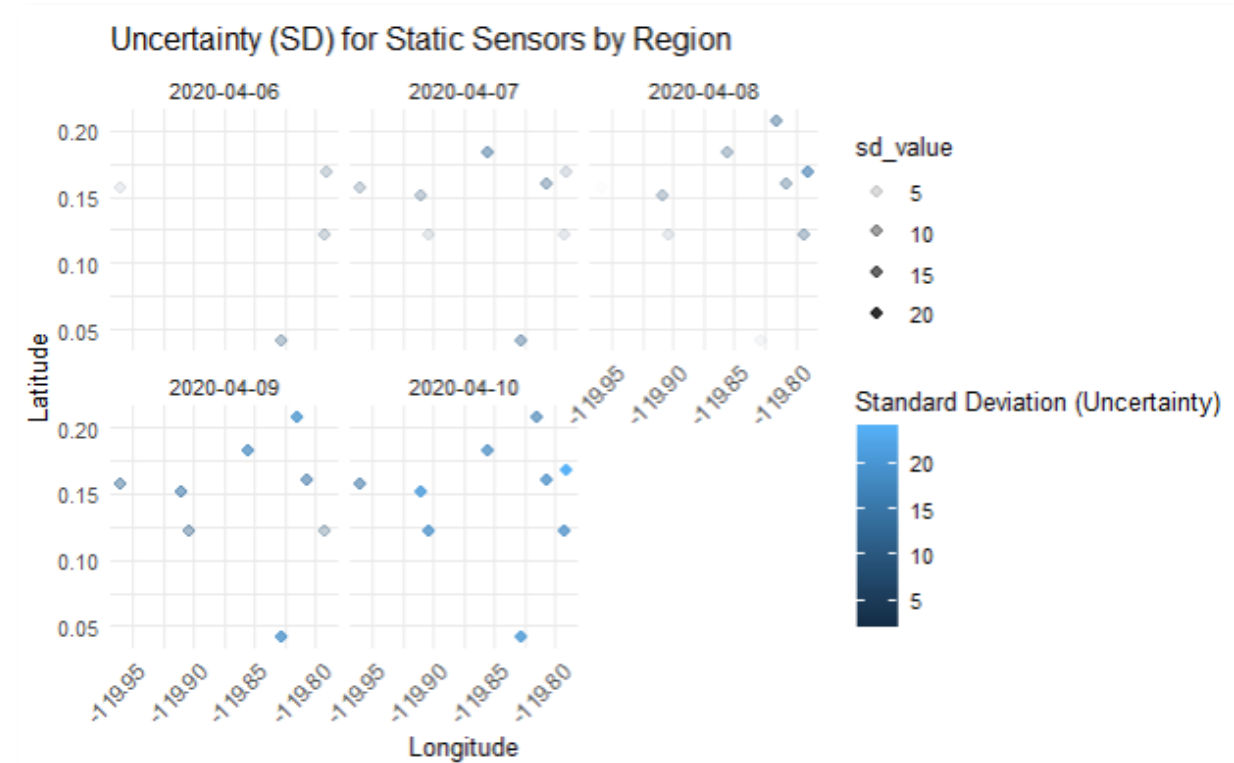
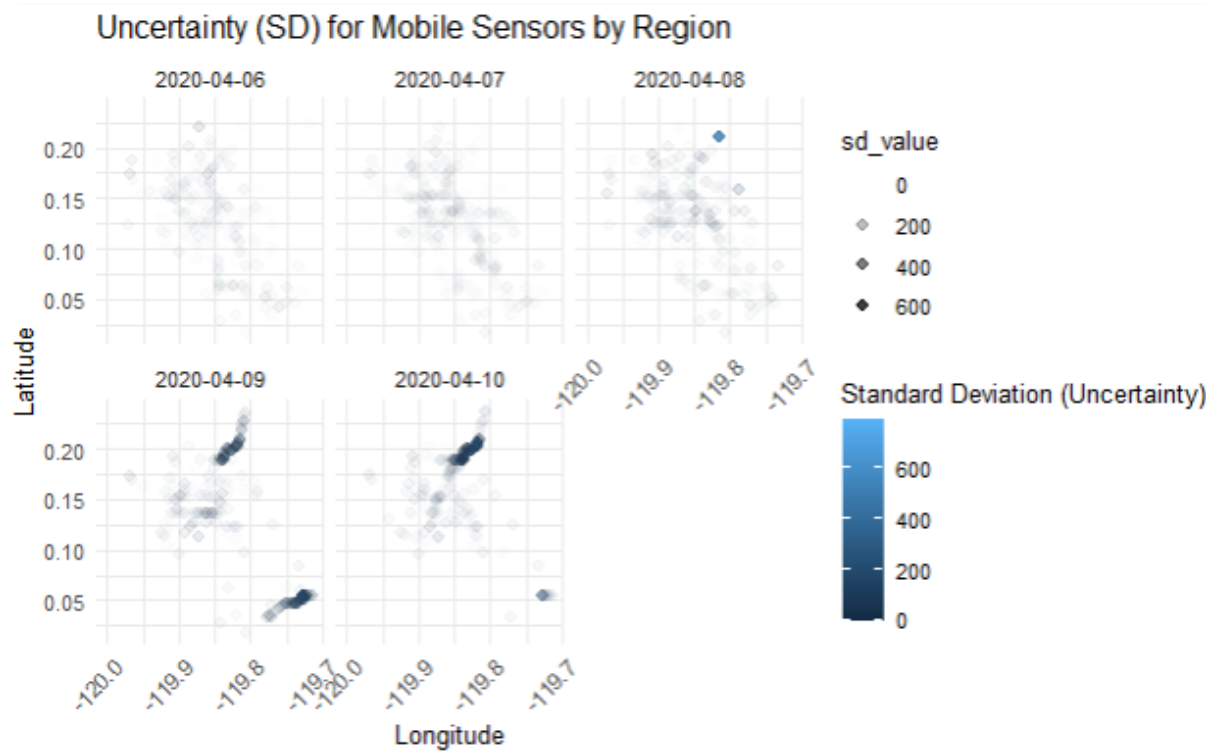


Visualization and Analysis: Uncertainty [Mean]

- Parametric T.test (p.value = 0.08) found moderate but non-significant signaling of a difference between daily averages of mobile and static sensors
- Non-parametric Wilcoxon Rank-Sum T.test (p.value=0.008) found significant difference between the daily averages between mobile sensor reports and static sensor reports



Visualization and Analysis: Uncertainty [SD]



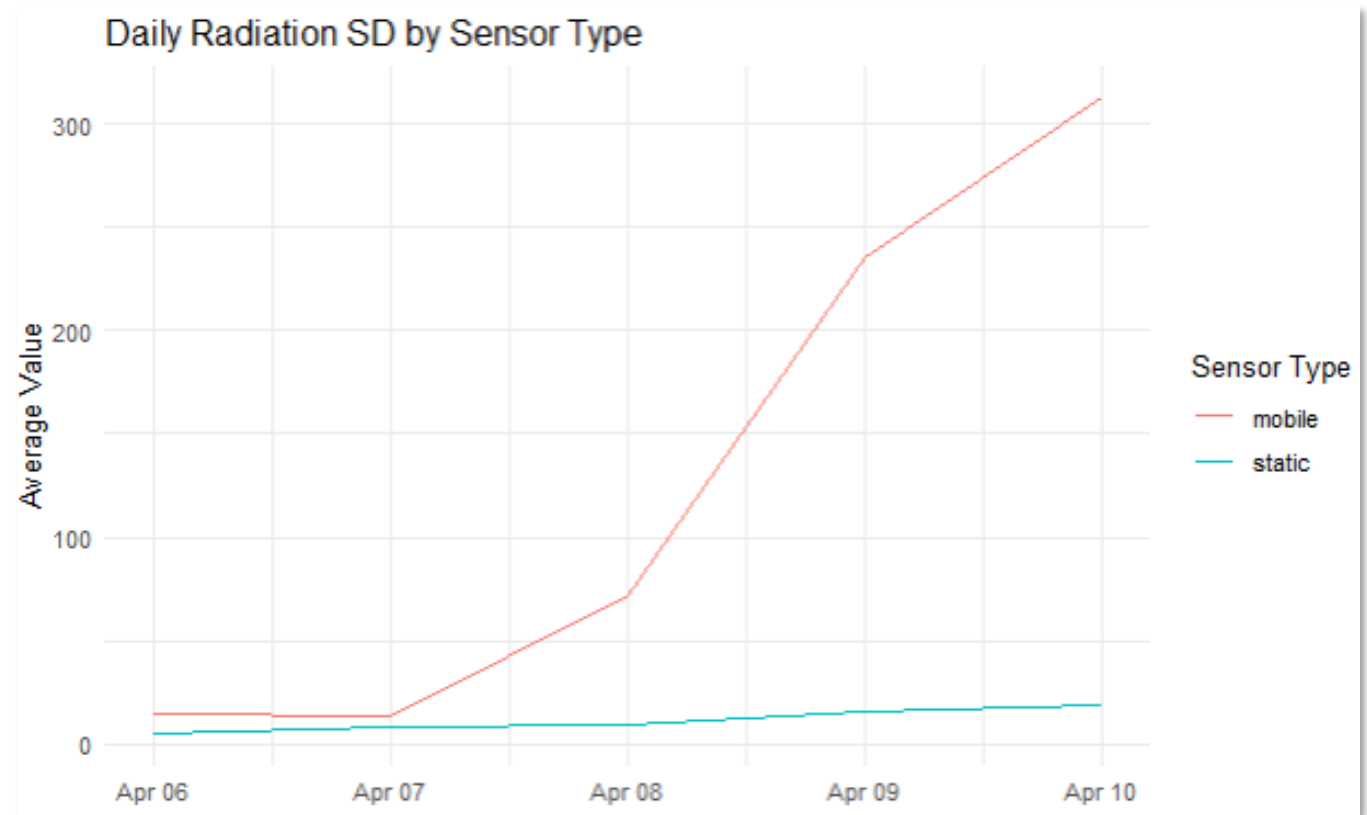
Concentrations of Variance where in relatively the same geographical locations between mobile and static sensor reports



Visualization and Analysis: Uncertainty [SD]



Standard Variation had significant change on Wednesday for mobile users, but static sensors so nominal changes in variation.

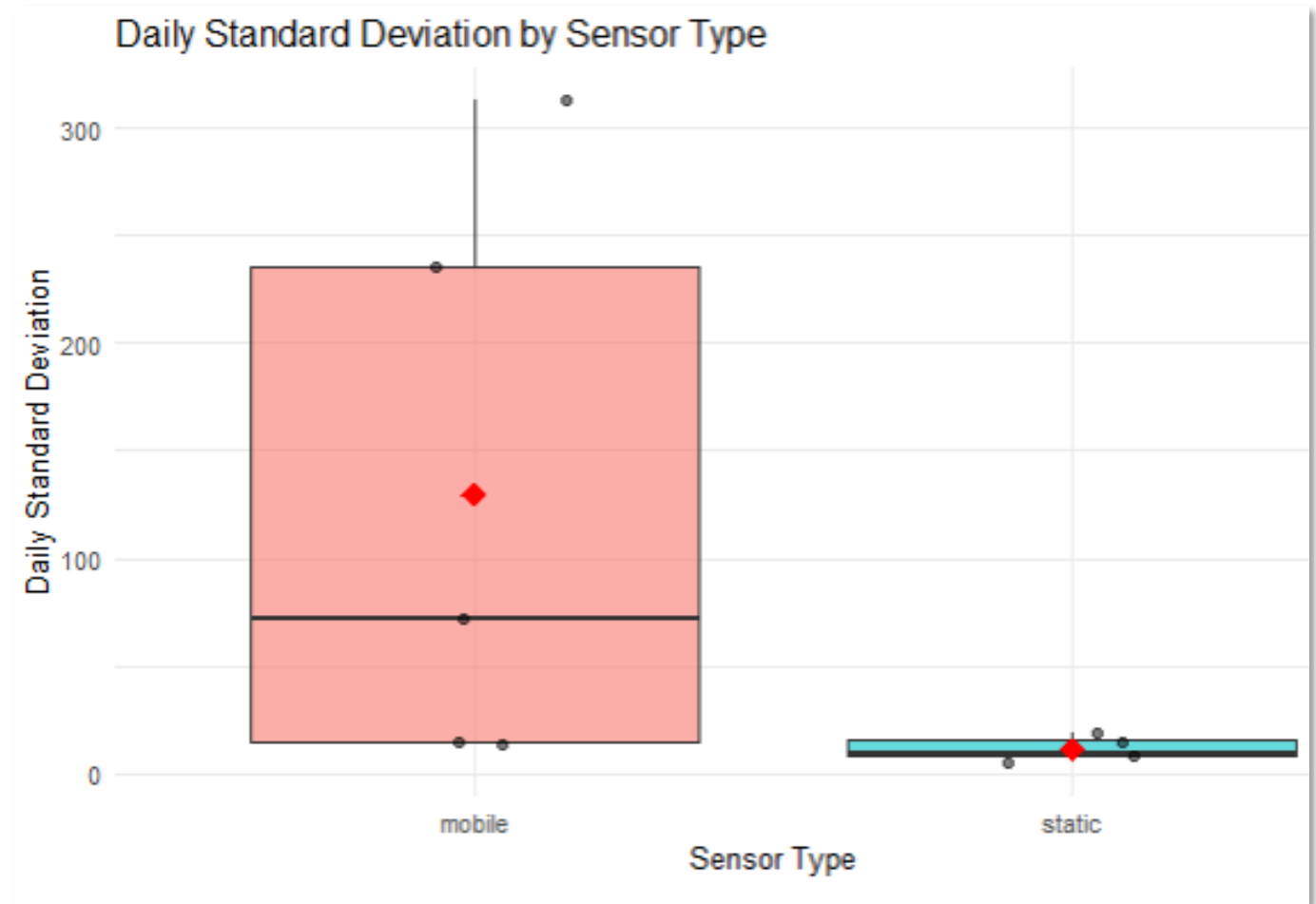


Visualization and Analysis: Uncertainty [SD]

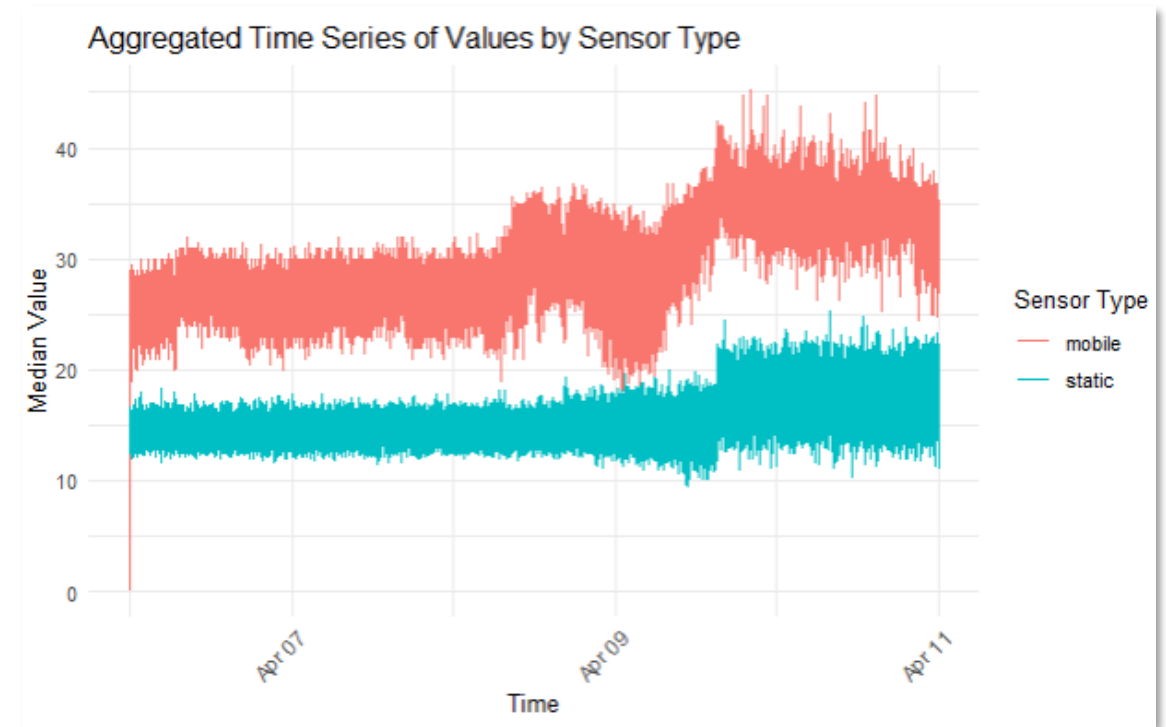
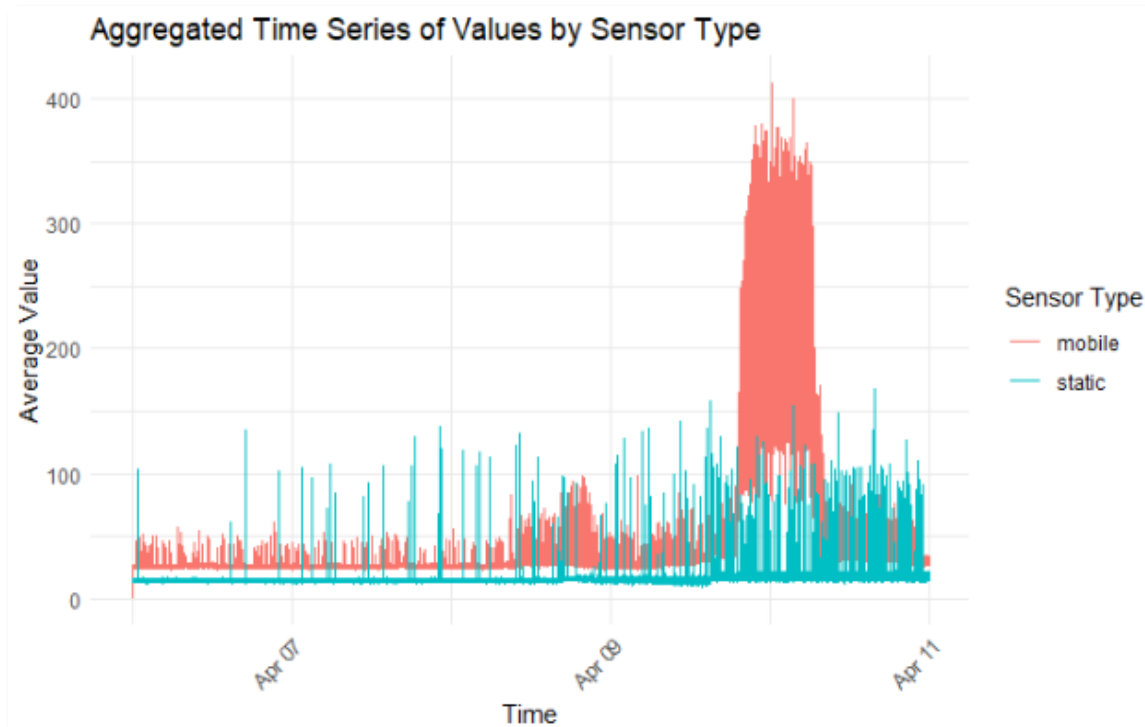


Neither parametric T.test (p.value = 0.12) or Non-parametric Welch T.test (0.095) found significant difference between the daily variation between mobile sensor reports and static sensor reports

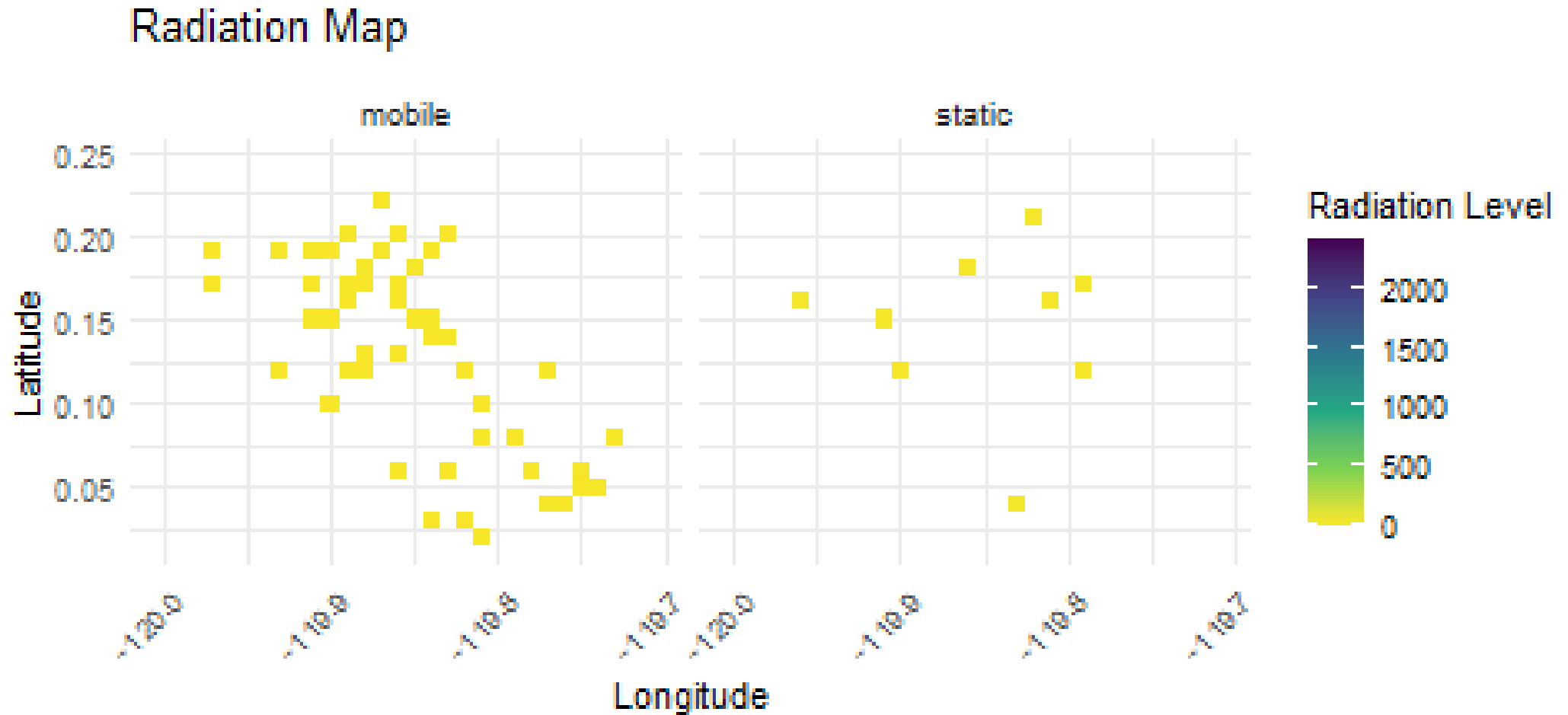
Descriptively, more variability (mobile group sd average~130, static group sd average~11)



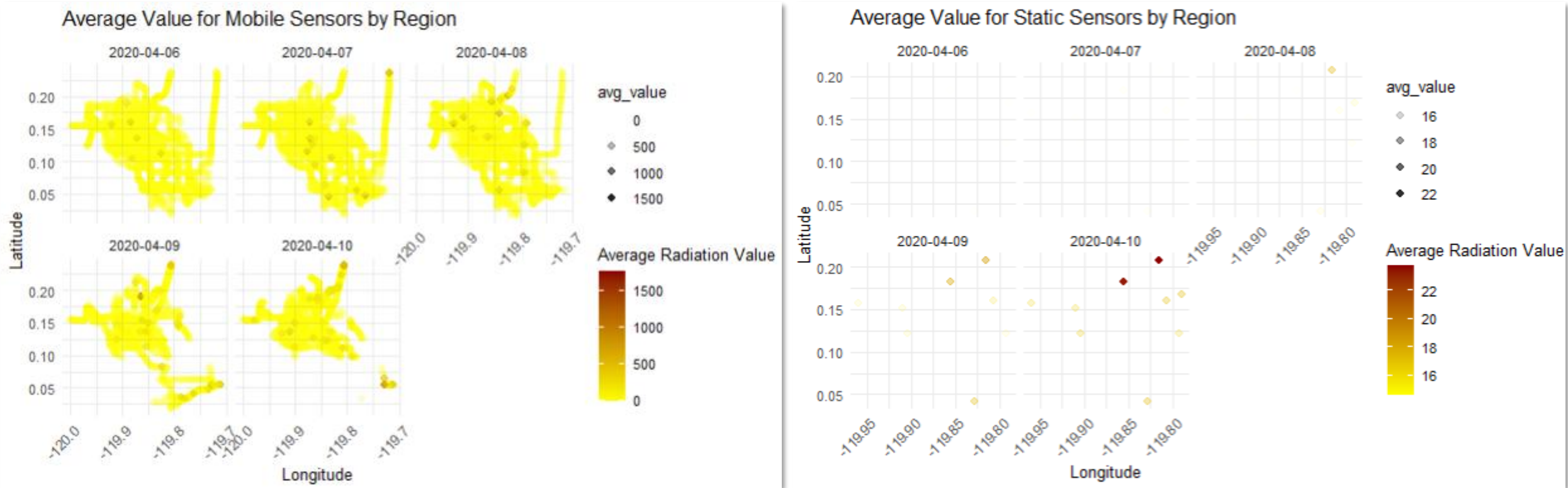
Visualization and Analysis: Troubled Area [Time]



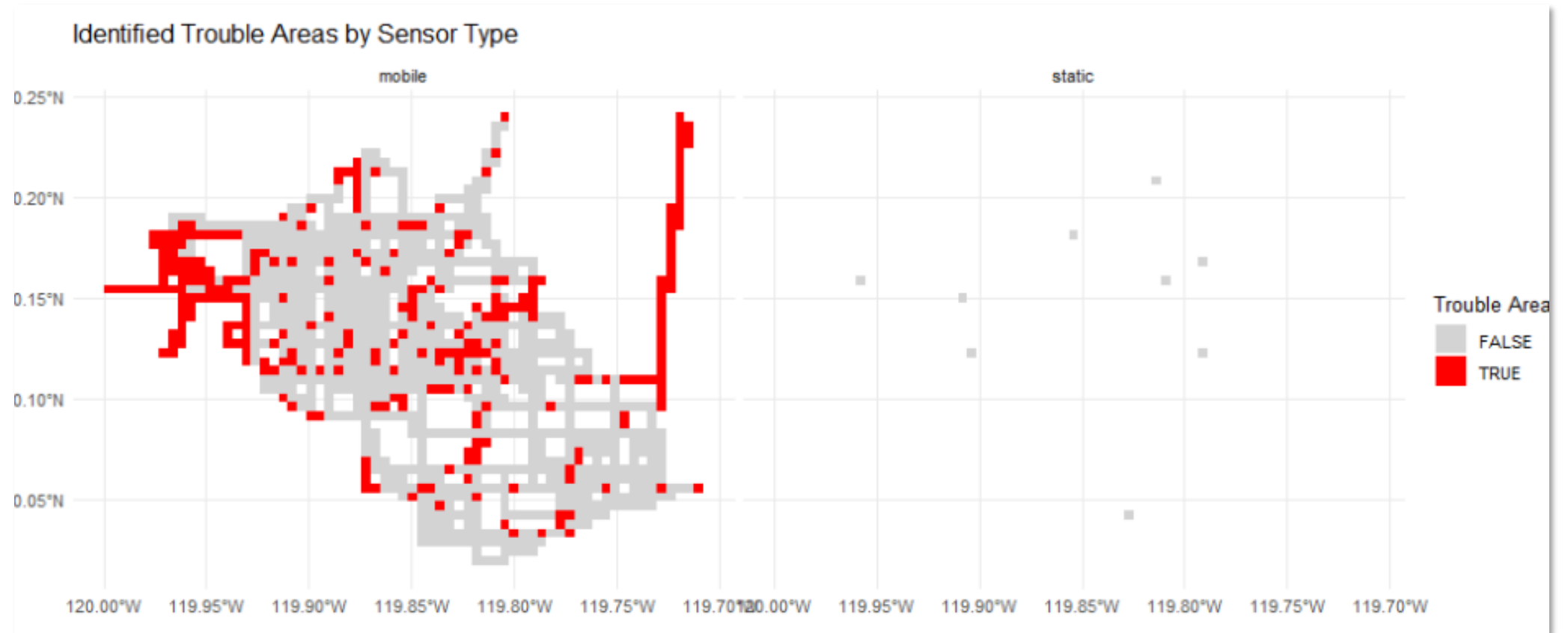
Visualization and Analysis: Troubled Area



Visualization and Analysis: Troubled Area



Visualization and Analysis: Troubled Area

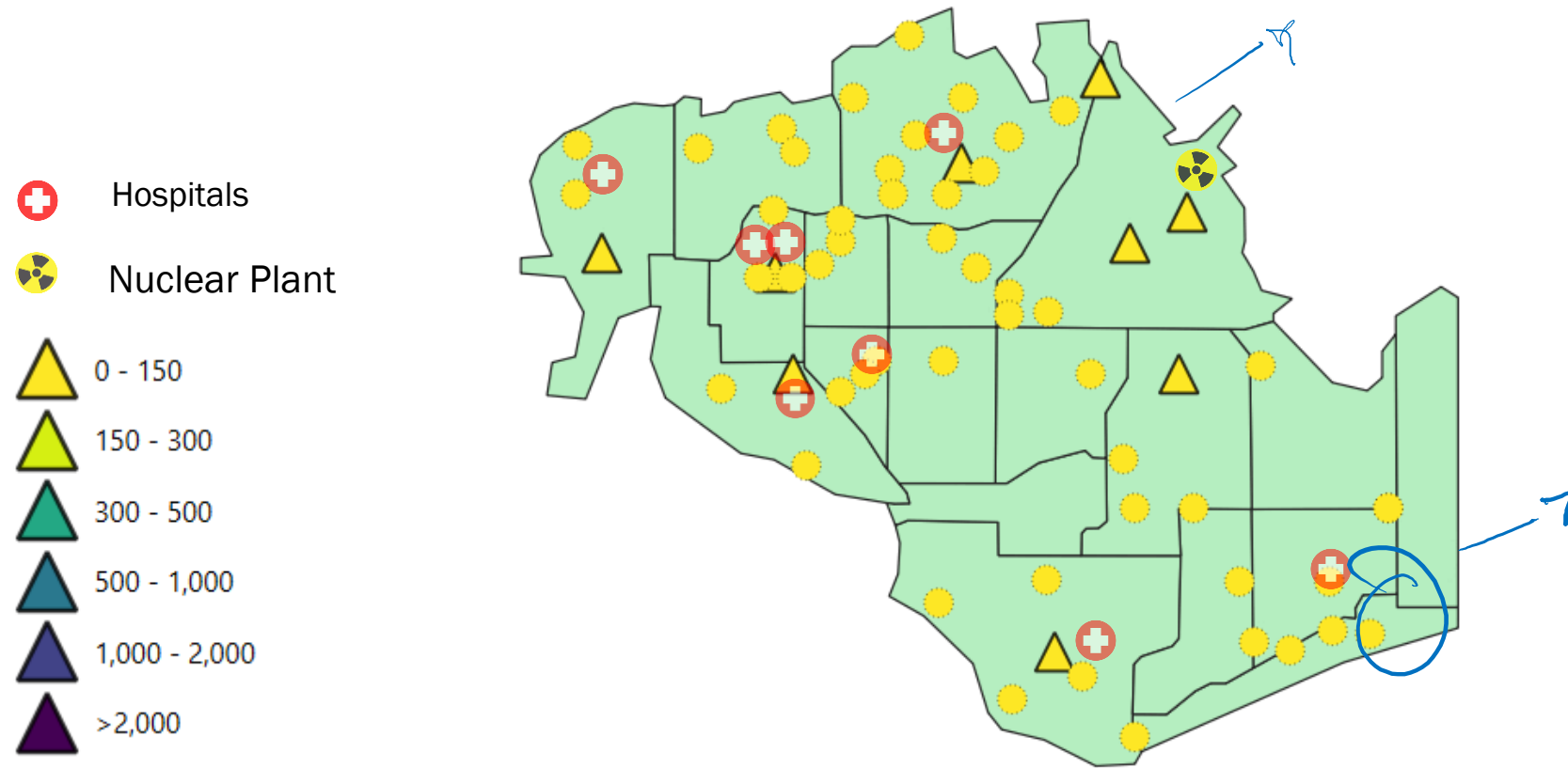


The **TRUE**/FALSE values in the trouble_area column represent whether each grid cell **exceeds the defined threshold** (e.g., the 75th percentile of mean_value)



Data Visualization and Analysis: Animation

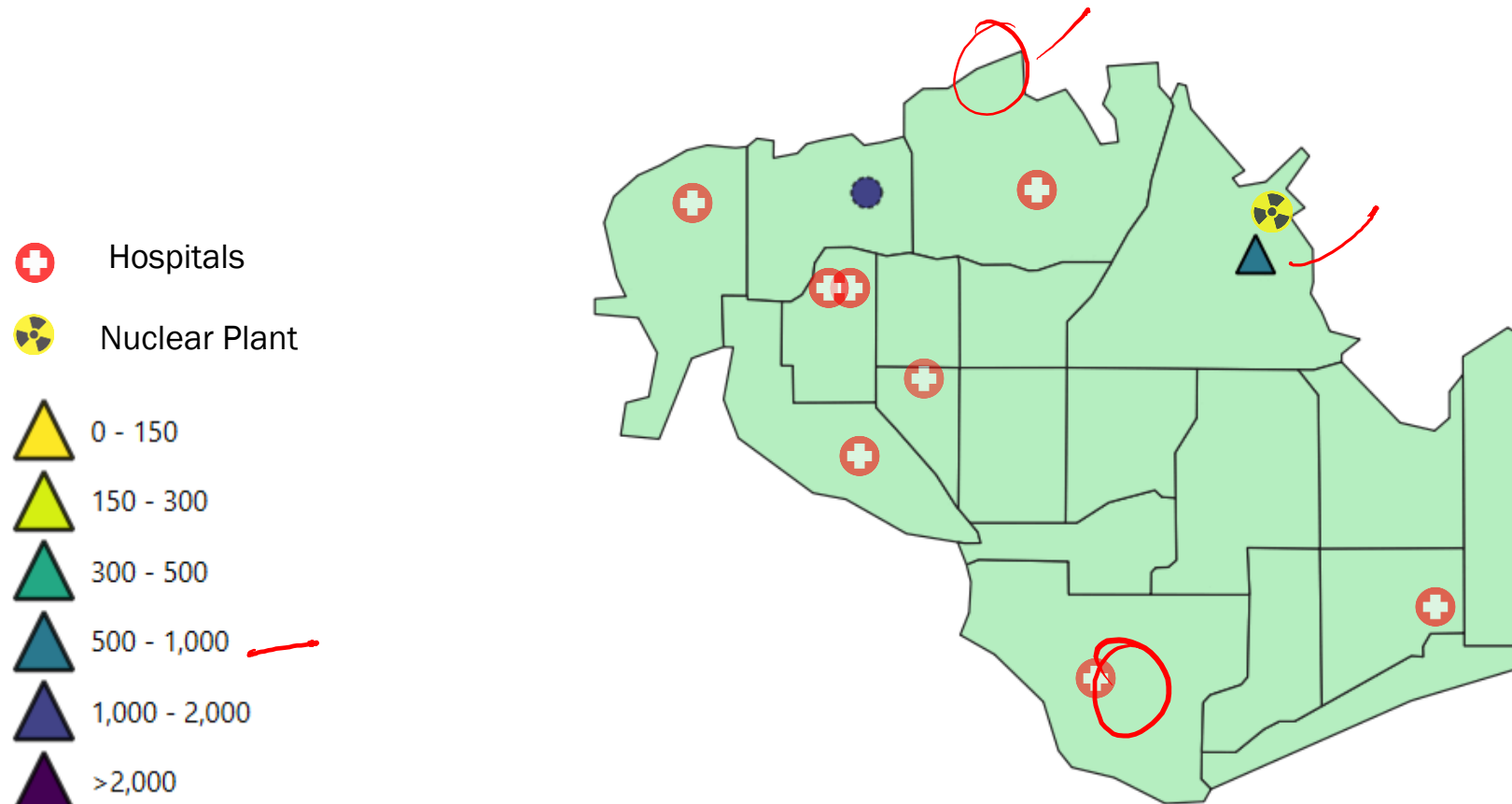
The following is the animation shows Static (triangle) and Mobile (circle) sensor readings from April 6th to 10th.



2020-04-06-00

Data Visualization and Analysis: Animation

The following animation shows Static (triangle) and Mobile (circle) sensor readings from April 6th to 10th. Radiations above 150 CPM [1]

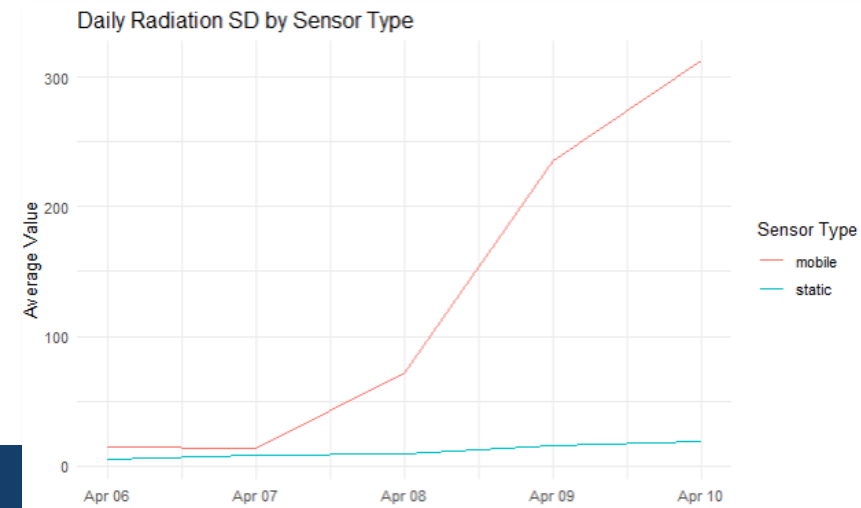
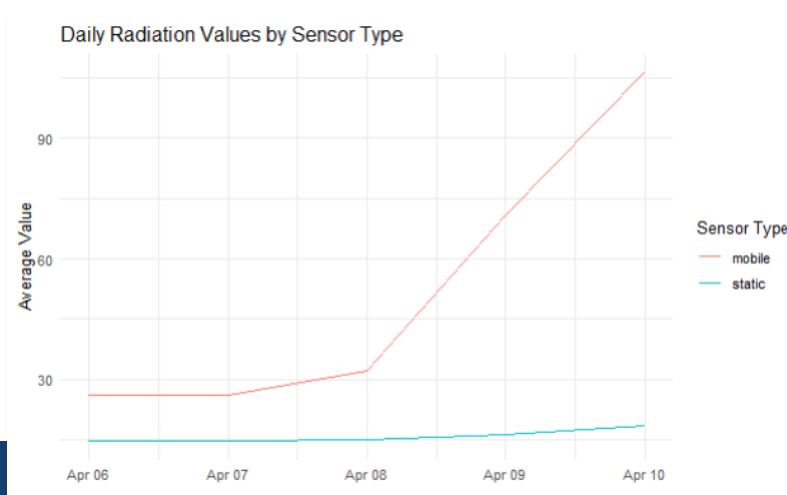


2020-04-06-00



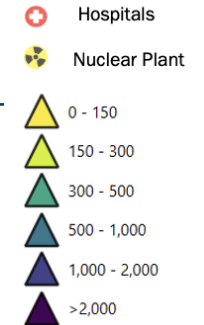
Conclusion: Mobile vs. Static Readings

- Daily averages between mobile and static readings **differed significantly** with appropriate testing.
- **Fixed sensors** were calibrated and stationary, giving consistent, reliable readings
- **Mobile sensors** varied in build quality and calibration, leading to inconsistent data
- Vehicle-based sensors were affected by **contamination and shielding**
- Some cars were directly contaminated, falsely elevating radiation readings
- **Movement and human error** introduced additional noise and uncertainty

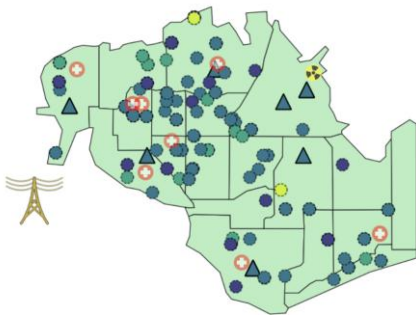




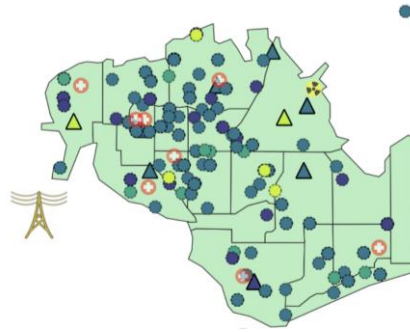
Conclusion: Plans for Cleanup



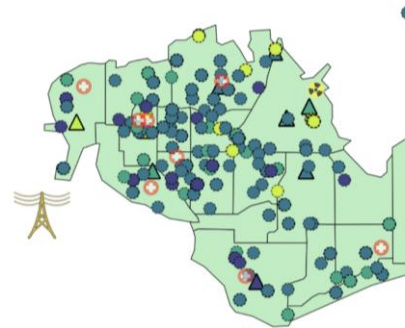
4/7: 3 PM



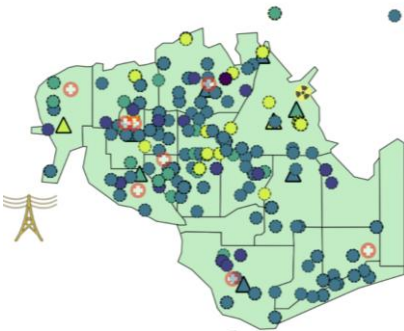
4/8: 9 AM



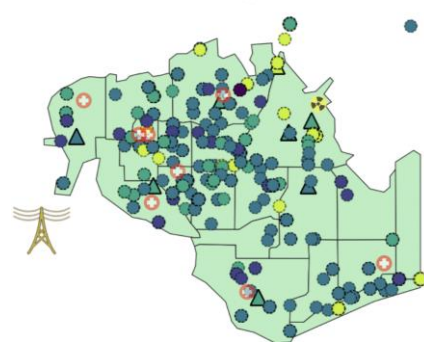
4/8: 10 PM



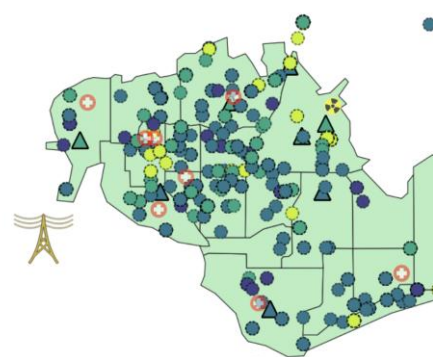
4/9: 8 AM



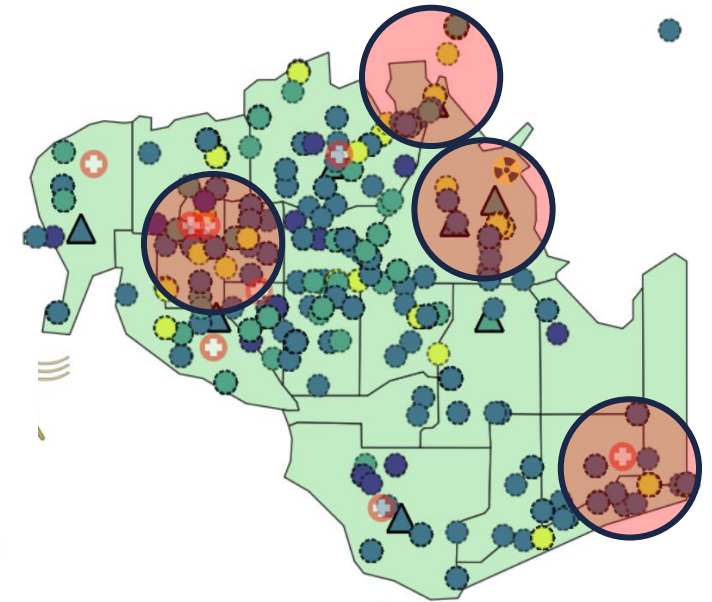
4/9: 8 PM



4/10: 1 AM



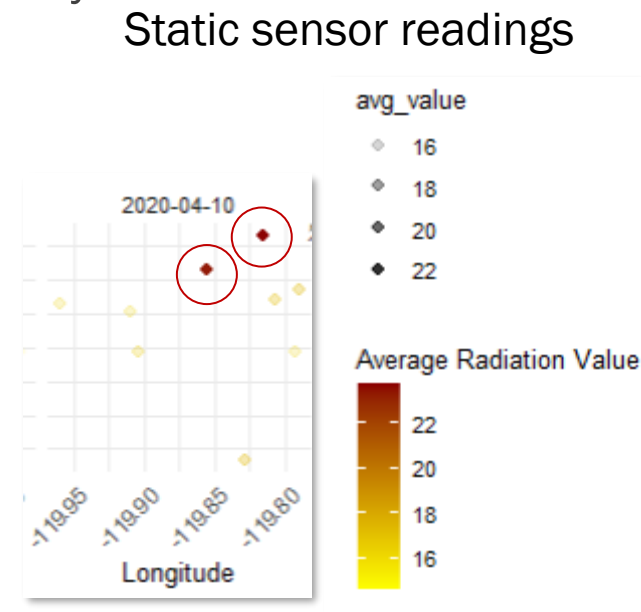
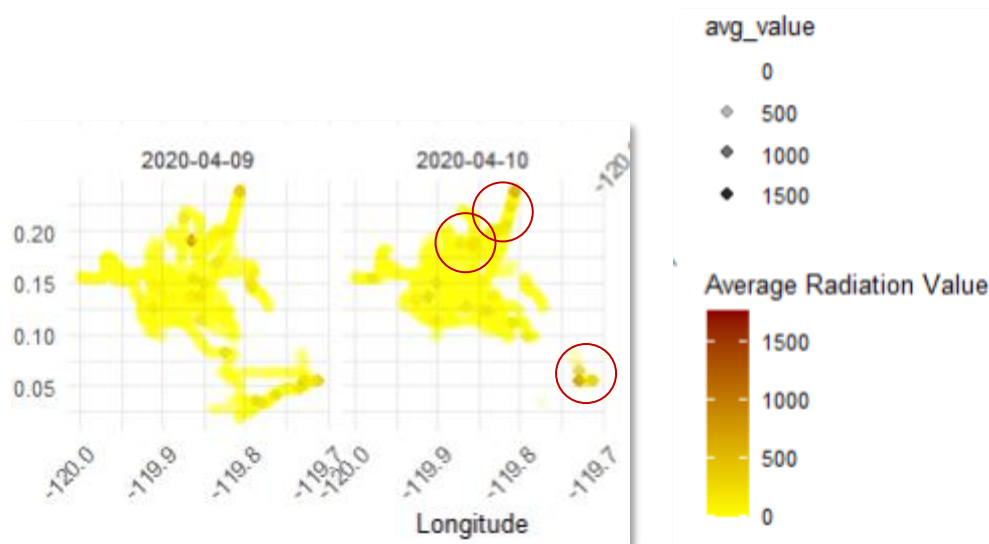
4/10: 10 PM





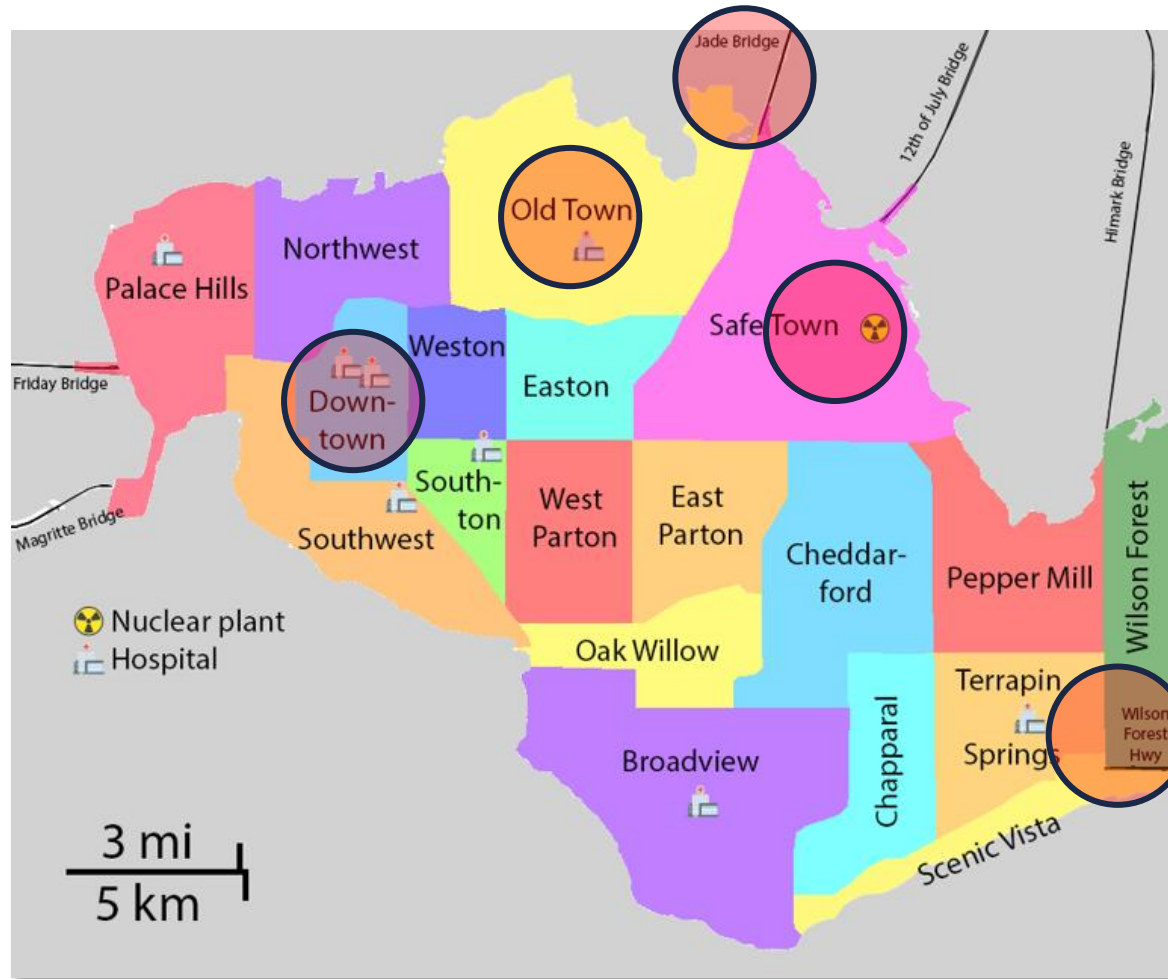
Conclusion: Plans for Cleanup

- Consistent high contamination found near **Jade Bridge** and **Old Town Hospital** across both sensor types
- **Mobile sensors** reveal additional high radiation near **Wilson Forest Hwy**, not detected by static sensors
- **Mobile network adds spatial insight**, despite higher uncertainty





Conclusion: Plans for Cleanup





Future Work

- **Spatial pairing of sensor data** not performed; no direct location matches between mobile and static sensors
- Future analysis could use:
 - **Clustering** to group readings by geographic zones
 - **Nearest point analysis** to compare static and mobile readings in proximity
- Incorporate **baseline/background radiation levels** for better thresholding
- Add **temporal clustering** to detect persistent vs. transient radiation patterns
- Consider **sensor confidence scoring** to account for device reliability

